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show your work!: a workflow for open source scientific articles

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ABSTRACT

This paper introduces `showyourwork`, a workflow that enables the creation and distribution of fully reproducible and open source scientific articles.

1. INTRODUCTION

As astronomical research software becomes increasingly more complex, and as research results become increasingly more interdependent, it becomes ever more challenging to ensure the validity and correctness of results published in the literature. Unfortunately, the current peer review system in astronomy is simply not set up to do this. Checking all of the results in a paper would require the painstaking and methodical review of all of the paper's methods—which usually means scrutinizing all of the code used to generate the figures, tables, and other quantities in the paper. In practice, this is virtually impossible for three reasons:

1. Modern codebases can be very large and often require deep familiarity with the software to use—not to mention review them. Volunteer referees rarely have the time to invest in learning new software in order to provide a comprehensive review.

2. Writing a paper in astronomy is rarely ever done in a linear, procedural fashion: the codebase is constantly changing, and the state of the code when (say) Figure 1 was produced may be very different from that when (say) Figure 2 was made. Moreover, many results depend on the execution of lengthy pipelines with intermediate steps, each potentially requiring manual tinkering that is not always documented and may be difficult to replicate exactly.

3. The majority of astronomical code is not open source and simply cannot be vetted by third parties. While there has been a marked increase in the number of open source astronomical tools in recent years (e.g., `astropy`, `exoplanet`, `emcee`, `exofast`...), most code associated with the generation of the results in individual papers is not open source; readers are often expected to take it on faith that there are no bugs in that code, or that the code works exactly as described in the text, with no pitfalls or missing details. Even when the code

is made publicly available, e.g., by being published on `GitHub`, it is often not documented sufficiently to enable one to execute it and reproduce the paper's results out-of-the-box. And even with proper documentation, the code may require external dependencies, custom virtual environments, or access to closed-source datasets that make it difficult or impossible for a third party to replicate it.

`showyourwork` was designed to tackle these three issues, making it easy to develop, publish, and distribute truly open and reproducible research papers in astronomy and other scientific disciplines. At its core, it is a command-line tool that builds papers from a set of instructions contained in a `GitHub` repository and organized into `TeX` files, figure scripts, pipeline files, and configuration/specifications files. Every time the user pushes a new commit to `GitHub`, the article is automatically built on the cloud using `GitHub Actions` and the resulting PDF is pushed to a separate branch of the repository. The build step—which sets up the conda environment, generates all figures from scratch (with intelligent caching), and compiles the PDF—acts as a unit test for the paper. If it passes, the paper is (by definition) reproducible.

`showyourwork` works out of the box for simple projects, in which each figure is generated by running a given script. But it also works for more complicated pipelines, such as projects that depend on many intermediate steps or those that require running expensive simulations on clusters. The workflow interfaces directly with `Zenodo`, allowing users to automatically upload the results of simulations so that expensive build steps can be bypassed on the cloud. In fact, most of the features under the hood are there to make the workflow as flexible and customizable as possible.

Papers that use `showyourwork` can be reproduced by cloning the associated repository and running `showyourwork`. Such papers (like this one!) include clickable icons next to each of their figures linking to (1) the exact version of the script on `GitHub` used to generate them and (2) the exact version(s) of the `Zenodo`-hosted dataset(s) used in their creation.

2. USING `showyourwork`

In this section we provide basic, high-level instructions on how to install and use `showyourwork`. These instructions are for version `0.3.0` of the code, the current version at the time of writing (July 2022) and are subject to change in future releases. For more details, including custom commands, settings, examples, and troubleshooting tops, please refer to the documentation for the specific version of `showyourwork` at https://show-your.work.

2.1. *Prerequisites*

`showyourwork` requires the `conda` package manager[1] and is currently tested only on Unix-like operating systems (such as Linux, Ubuntu, or MacOS). Users must also

[1] https://www.anaconda.com/products/distribution

have access to a GitHub account; other git platforms are not currently supported. Note that users do *not* need a local installation of a TeX distribution, as showyourwork uses the conda-managed tectonic package to compile articles.

2.2. *Installation*

showyourwork can be installed with the Python package manager pip:

```
python -m pip install -U showyourwork
```

(recommended) or from source on GitHub:

```
git clone https://github.com/showyourwork/showyourwork
cd showyourwork
python -m pip install .
```

2.3. *Reproducing an article*

Any project based on showyourwork can be reproduced by cloning its GitHub repository and running showyourwork. For example, to reproduce this paper, run:

```
git clone https://github.com/showyourwork/showyourwork-paper
cd showyourwork-paper
showyourwork
```

This will set up a custom conda environment for the workflow, download the required datasets from Zenodo, build all of the figures, and generate a PDF identical to this one.

2.4. *Creating a new article*

New projects may be created by running

```
showyourwork setup user/repo
```

where user and repo are the user's GitHub handle and the repository name, respectively. After the user answers a few prompts, showyourwork will set up a local git repository repo with the correct structure and placeholder files (see §2.5). The article may then be built by running

```
showyourwork
```

from the root of the repository.

2.5. *Repository structure*

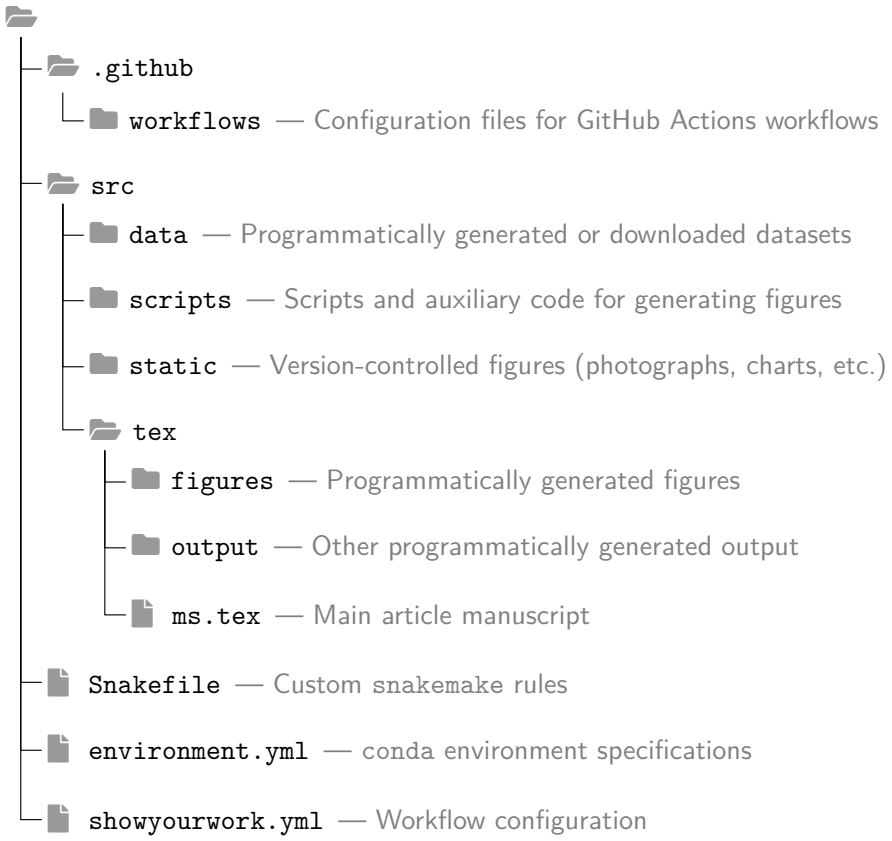

Figure 1. The basic repository structure for an open source scientific article built with `showyourwork`. This figure was automatically generated from the `TikZ` code in `src/scripts/tree.tex` by specifying a custom command in `showyourwork.yml`.

Figure 1 shows the basic directory structure for a `showyourwork` article repository as of version `0.3.0`. Repositories are comprised of a configuration files at the root level and a directory called `src` containing the manuscript files and all of the the code and workflow scripts needed to render the final article PDF.

3. HOW IT WORKS

Coming soon.

4. EXAMPLES

In this section, we showcase a few projects based on `showyourwork` to illustrate how the workflow can facilitate the development and dissemination of open source scientific articles.

The most fundamental feature of `showyourwork` is the automatic generation of the figures in a paper, such as Figure 2, which we reproduce from Wagg et al. (2022). This figure was generated from the script `src/scripts/eccentricity.py` and has only a single other dependency: the `conda` environment file `environment.yml` (see Figure 3).

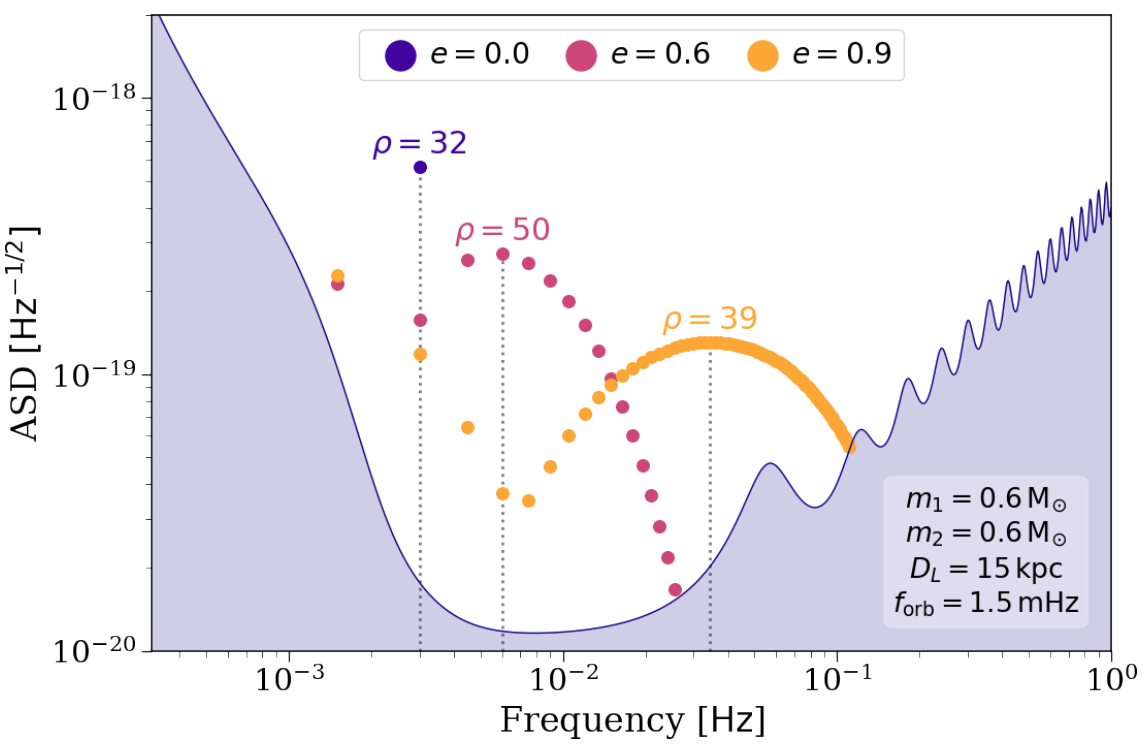

Figure 2. The effect of binary eccentricity on the detectability of a *LISA* gravitational wave source; reproduced from Figure 3 in Wagg et al. (2022). This figure was automatically generated from the script `src/scripts/eccentricity.py`.

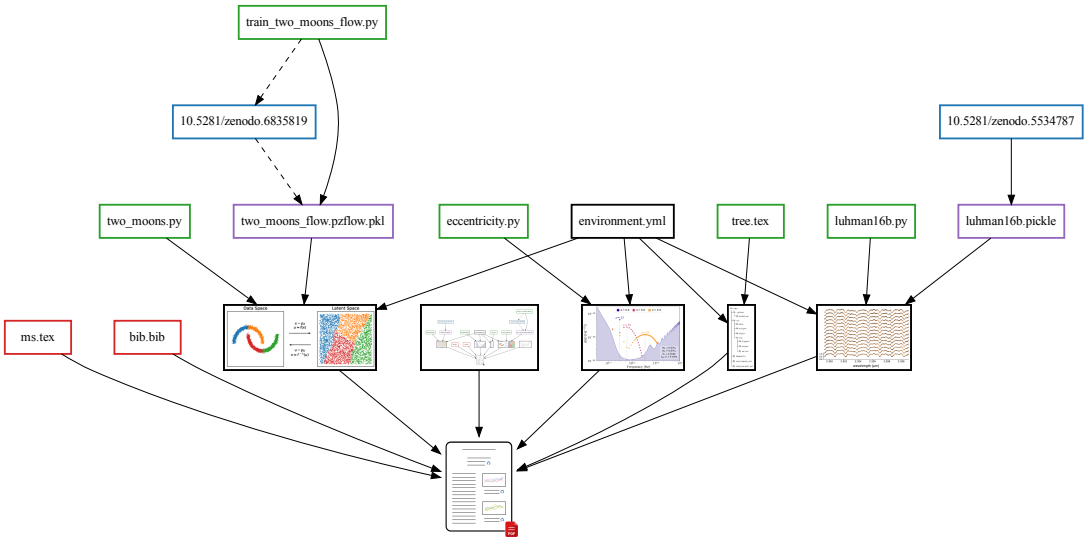

Figure 3. A directed acyclic graph (DAG) showing the complete list of dependencies for the article. Scripts are shown in green, `Zenodo` deposits in blue, datasets in purple, and `TeX` files in red. This figure is located in the `src/static` directory and, unlike the other figures in this article, is version controlled by `git`. The `src/static` directory is reserved for figures that are not programmatically generated and simply get copied over to the output directory at compile time.

REFERENCES

Wagg, T., Breivik, K., & de Mink, S. E. 2022, ApJS, 260, 52, doi: 10.3847/1538-4365/ac5c52